PARSNIP-NODE

Meshtastic LoRa mesh node, 915 MHz

Schematic

PAGE LIST

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11	Peripherals	GPIO expander, sensors, LEDs
12	Connectors	Expansion headers, e-Ink display

REVISION HISTORY

REV	DATE	DESCRIPTION OF CHANGES	BY
1.1	2026-10-03	Prototype schematic; PCB layout in progress	LimesKey

NOTES, UNLESS OTHERWISE SPECIFIED:

1. Resistance in ohms, capacitance in farads. All voltages are DC.
2. Resistors and capacitors are 0402.
3. DNP = do not populate. DNP symbols are drawn with a red X.
4. An overbar on a pin or net name denotes an active-low signal.
5. Labels with the same name are connected. Top-level sheets join through global labels and power symbols, subsheets of Root through sheet pins.
6. PWR_FLAG symbols are for ERC only and are not parts.
7. Some symbols are edited in the schematic. Do not run Tools > Update Symbols from Library.

LimesKey

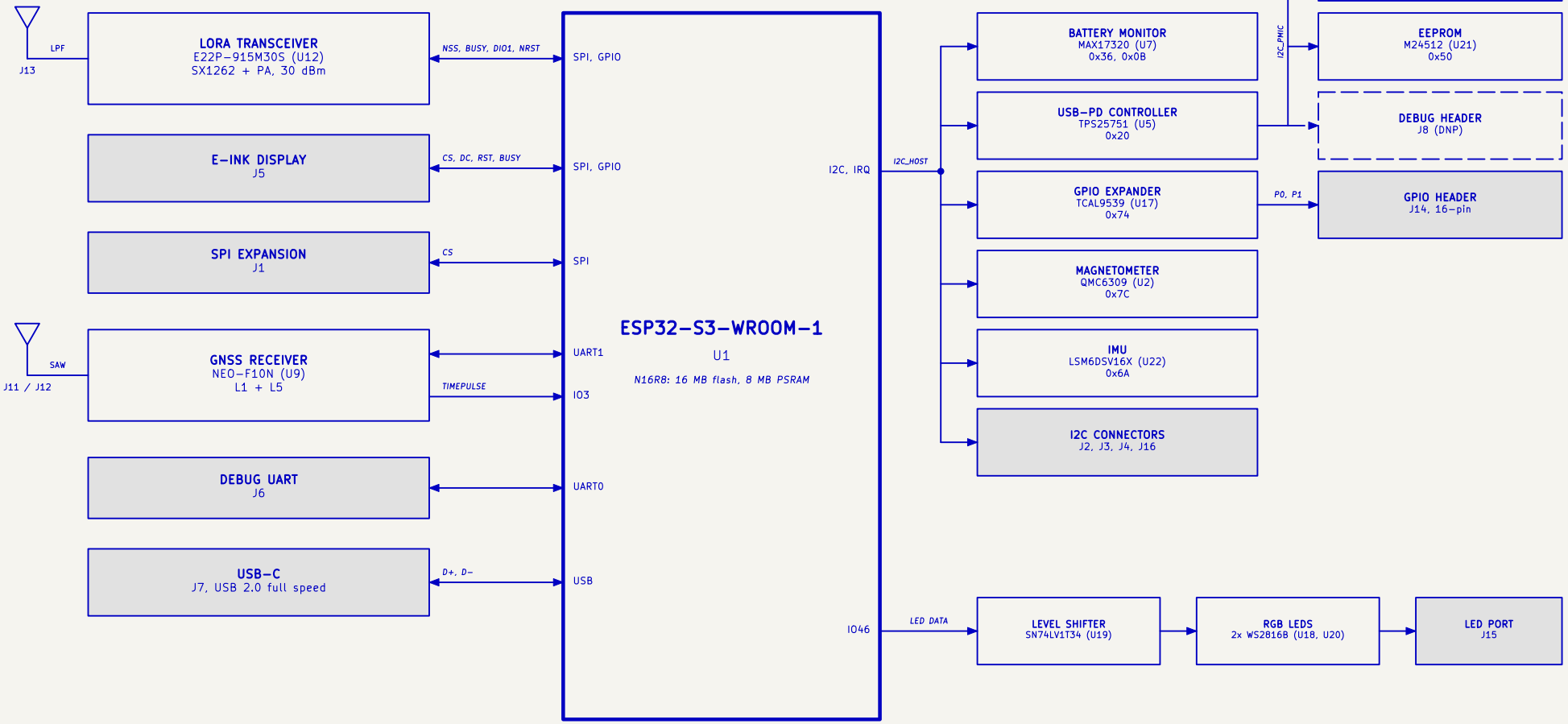
Sheet: /Overview/

File: overview.kicad_sch

Title: Cover

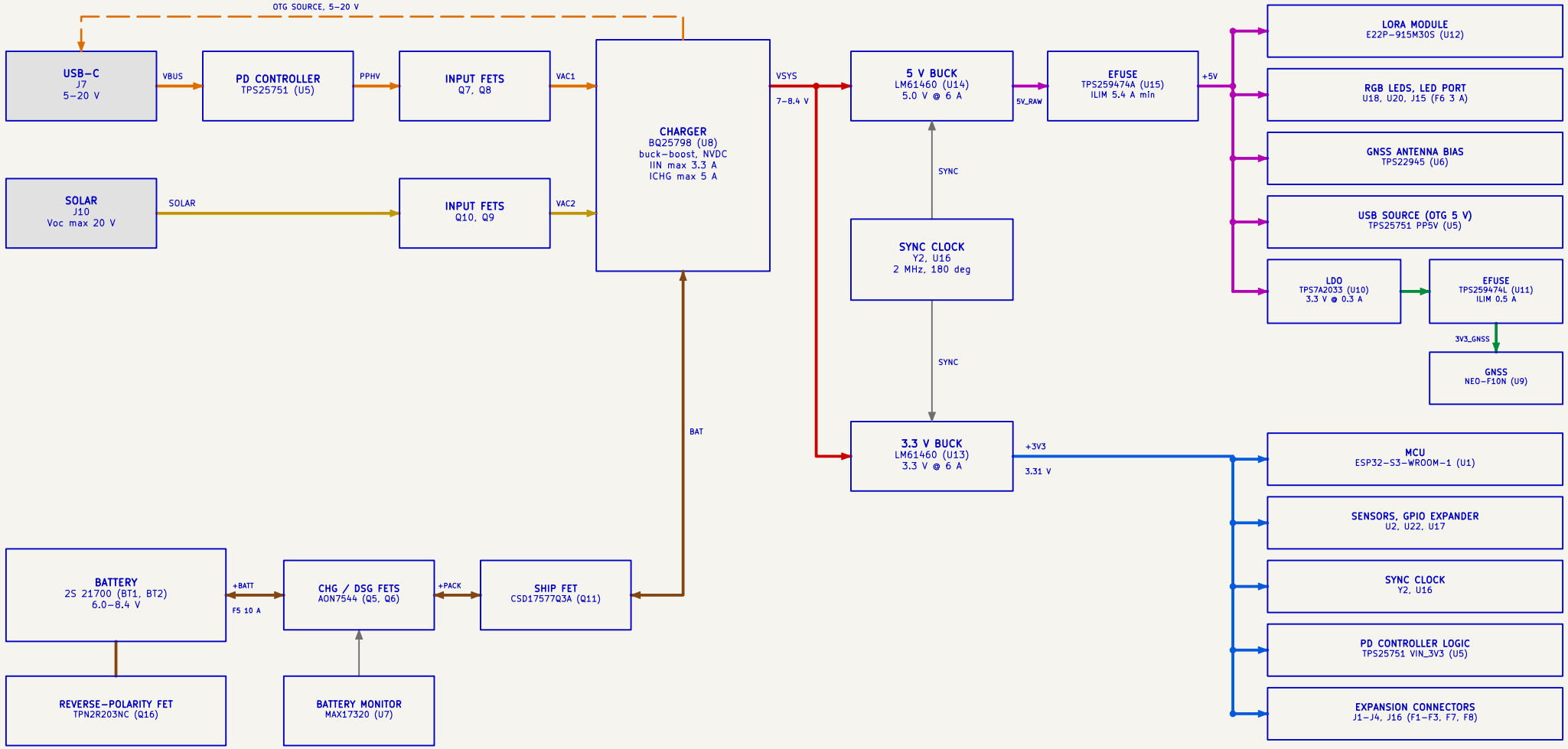
Size: A4	Date: 2026-10-03	**Rev: 1.1**
KiCad E.D.A. 10.99.0-5499-g0f5acecd00		Id: 1/12

BLOCK DIAGRAM

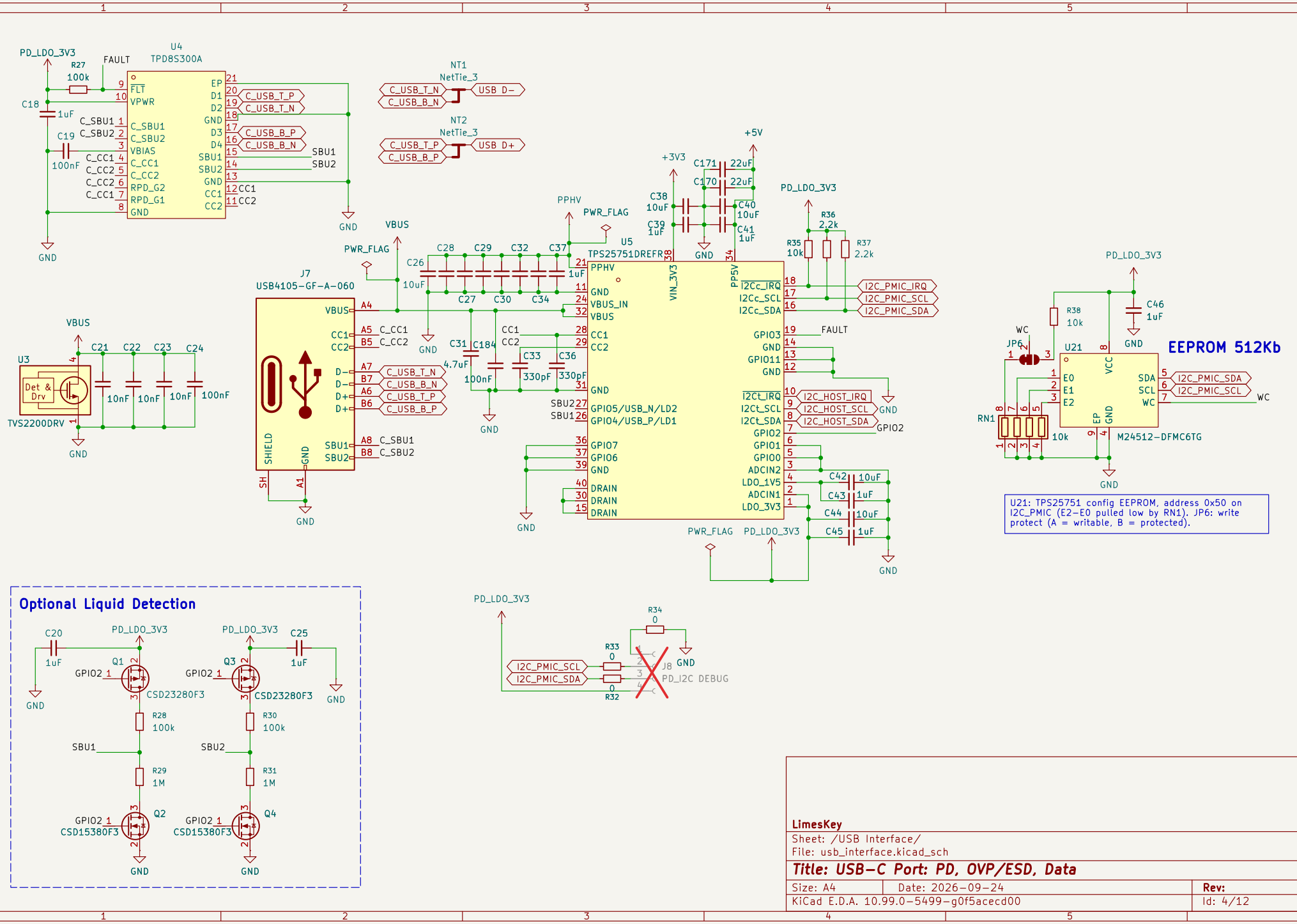


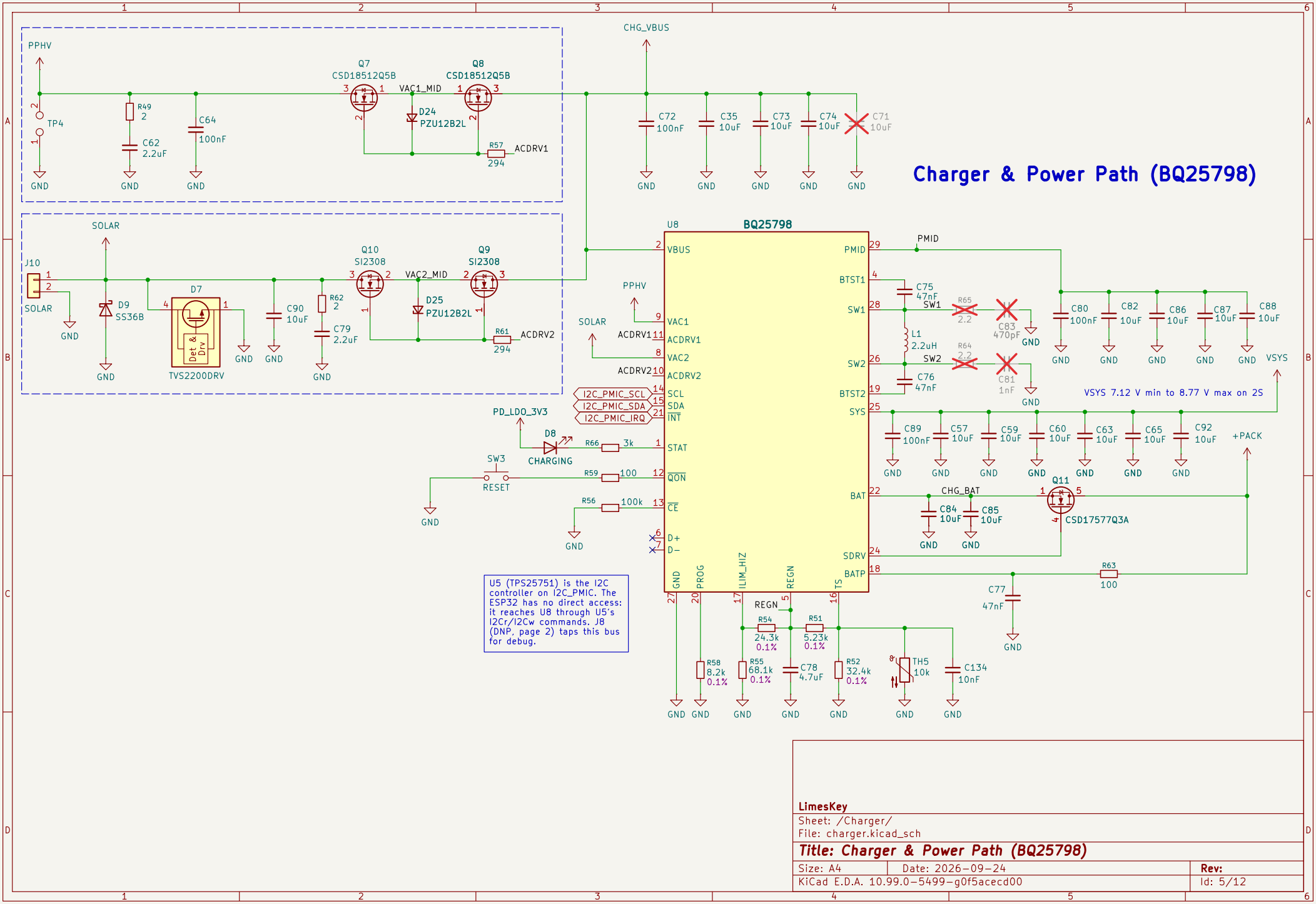
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Title: Block Diagram		
Size: A4	Date: 2026-10-03	**Rev: 1.1**
KiCad E.D.A. 10.99.0-5499-g0f5acecd00		Id: 2/12

POWER TREE

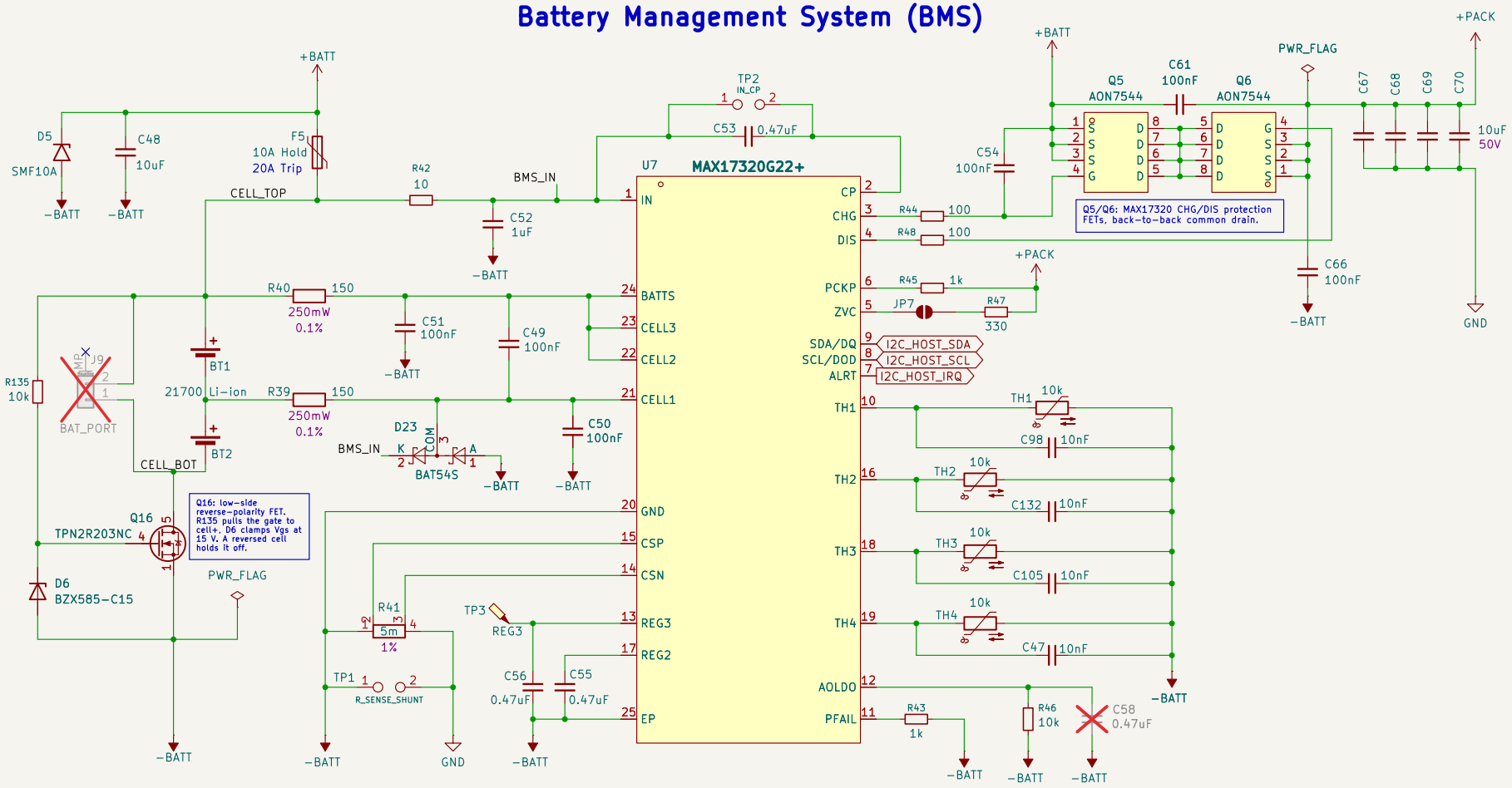


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Size: A4	Date: 2026-10-03	**Rev: 1.1**
KiCad E.D.A. 10.99.0-5499-g0f5acecd00		Id: 3/12

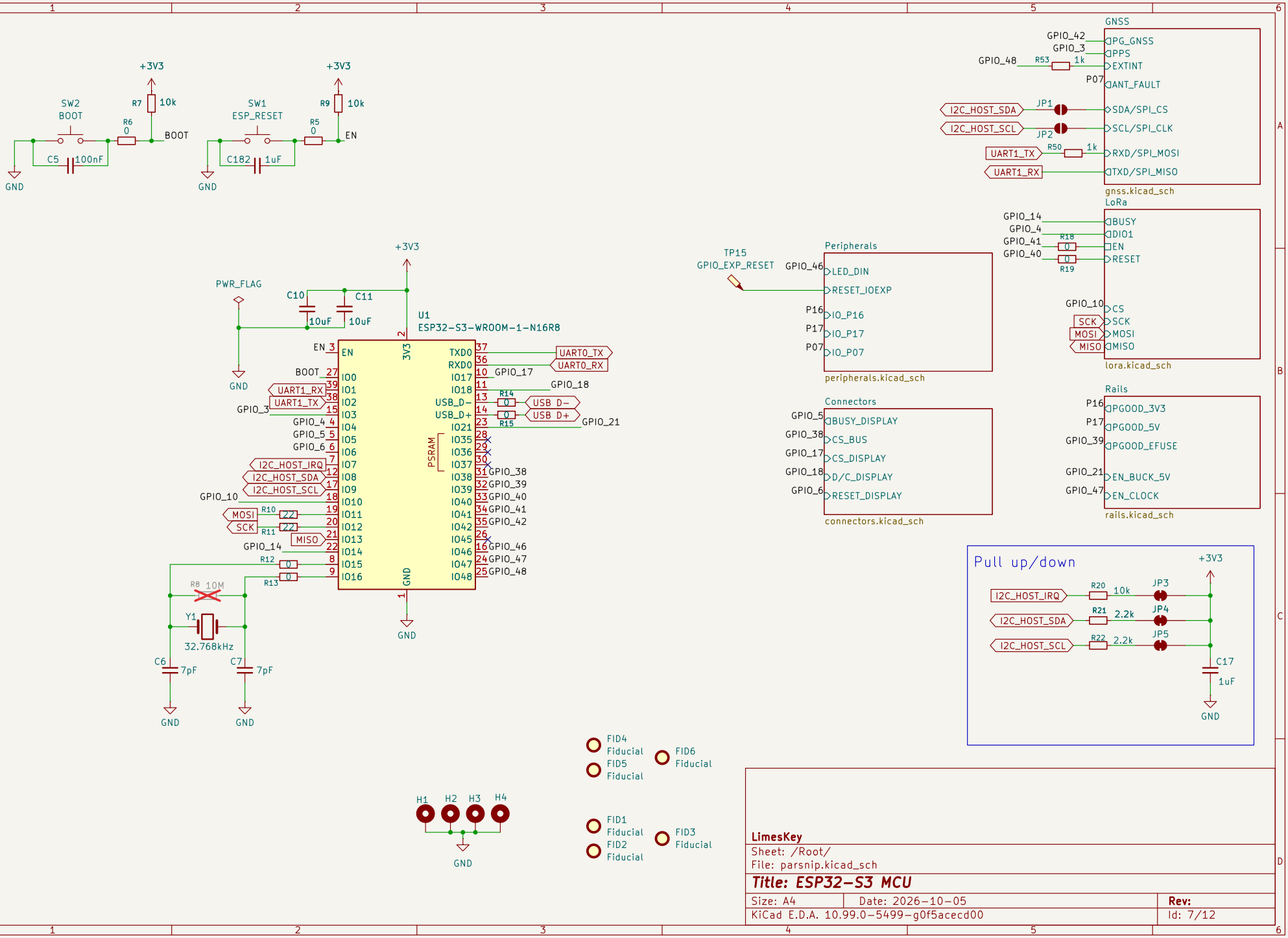

U4
TPD8S300A
PD_LDO_3V3
FAULT
R27
100k
C18
1uF
C19
100nF
C_SBU1
C_SBU2
C_CC1
C_CC2
RPD_G2
RPD_G1
C_USB_T_P
C_USB_T_N
C_USB_B_P
C_USB_B_N
SBU1
SBU2
CC1
CC2
GND
NT1
NetTie_3
USB D-
NT2
USB D+
U5
TPS25751DREFR
PPHV
PWR_FLAG
VBUS
+3V3
+5V
C171
22uF
C170
22uF
C38
10uF
C39
1uF
C40
10uF
C41
1uF
R36
2.2k
R35
10k
R37
2.2k
I2Cc_IRQ
I2Cc_SCL
I2Cc_SDA
I2C_PMIC_IRQ
I2C_PMIC_SCL
I2C_PMIC_SDA
J7
USB4105-GF-A-060
C26
10uF
C28
C29
C32
C37
1uF
C27
C30
C34
C31
4.7uF
C184
100nF
C33
330pF
C36
330pF
VIN_3V3
PP5V
GPIO3
GPIO11
I2Ct_IRQ
I2Ct_SCL
I2Ct_SDA
I2C_HOST_IRQ
I2C_HOST_SCL
I2C_HOST_SDA
GPIO2
GPIO1
GPIO0
ADCIN2
LDO_1V5
ADCIN1
LDO_3V3
GPIO5/USB_N/LD2
GPIO4/USB_P/LD1
GPIO7
GPIO6
DRAIN
U3
Det & Drv
TVS2200DRV
C21
C22
C23
C24
10nF
10nF
10nF
100nF
C42
10uF
C43
1uF
C44
10uF
C45
1uF
EEPROM 512Kb
U21
M24512-DFMC6TG
JP6
WC
R38
10k
C46
1uF
RN1
10k
E0
E1
E2
SDA
SCL
U21: TPS25751 config EEPROM, address 0x50 on I2C_PMIC (E2-E0 pulled low by RN1). JP6: write protect (A = writable, B = protected).
Optional Liquid Detection
C20
C25
1uF
Q1
Q2
Q3
Q4
CSD23280F3
CSD15380F3
GPIO2
R28
100k
R29
1M
R30
100k
R31
1M
SBU1
SBU2
R34
0
R33
0
R32
0
J8
PD_I2C DEBUG
LimesKey
Sheet: /USB Interface/
File: usb_interface.kicad_sch
Title: USB-C Port: PD, OVP/ESD, Data
Size: A4
Date: 2026-09-24
Rev:
KiCad E.D.A. 10.99.0-5499-g0f5acecd00
Id: 4/12

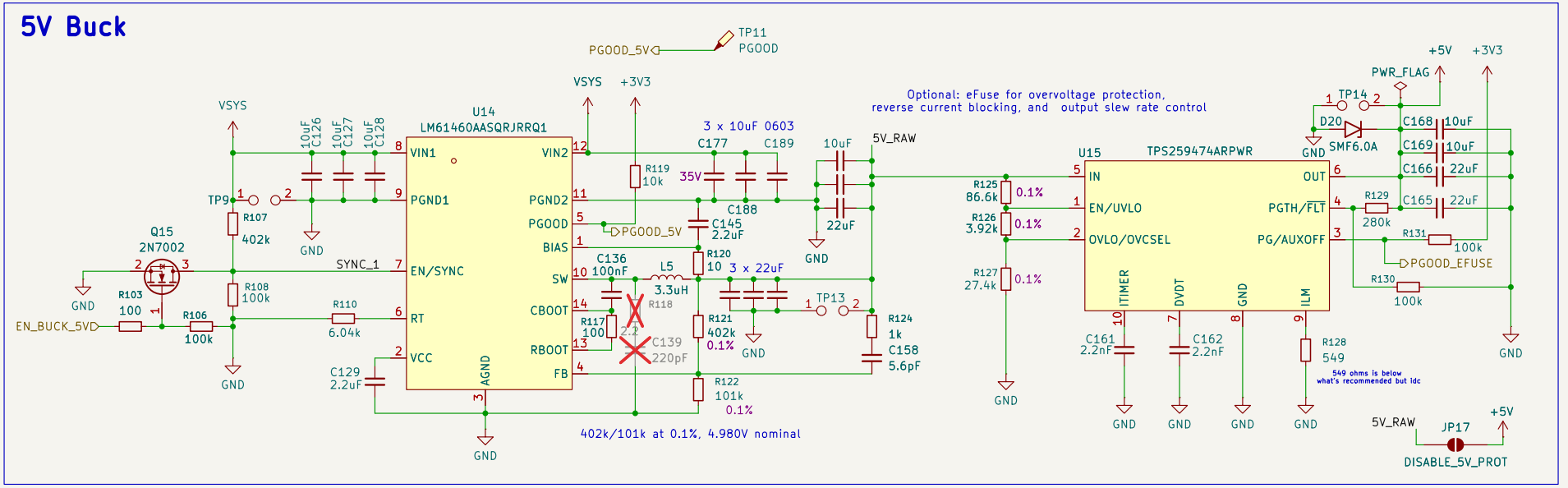

Charger & Power Path (BQ25798)
U8
BQ25798
PPHV
CHG_VBUS
SOLAR
PMID
VSYS
+PACK
VSYS 7.12 V min to 8.77 V max on 2S
PD_LDO_3V3
CHG_BAT
VAC1_MID
VAC2_MID
ACDRV1
ACDRV2
I2C_PMIC_SCL
I2C_PMIC_SDA
I2C_PMIC_IRQ
REGN
U5 (TPS25751) is the I2C controller on I2C_PMIC. The ESP32 has no direct access: it reaches U8 through U5's I2Cr/I2Cw commands. J8 (DNP, page 2) taps this bus for debug.
LimesKey
Sheet: /Charger/
File: charger.kicad_sch
Title: Charger & Power Path (BQ25798)
Size: A4
Date: 2026-09-24
Rev:
KiCad E.D.A. 10.99.0-5499-g0f5acecd00
Id: 5/12

Battery Management System (BMS)



LimesKey		
Sheet: /BMS/		
File: bms.kicad_sch		
Title: Battery Management System (BMS)		
Size: A4	Date: 2026-09-27	Rev:
KiCad E.D.A. 10.99.0-5499-g0f5acecd00		Id: 6/12


SW2
BOOT
R7 10k
R6 0
BOOT
C5 100nF
GND
+3V3
SW1
ESP_RESET
R9 10k
R5 0
EN
C182 1uF
GND
+3V3
PWR_FLAG
C10 10uF
C11 10uF
GND
U1
ESP32-S3-WROOM-1-N16R8
EN
BOOT
UART1_RX
UART1_TX
GPIO_3
GPIO_4
GPIO_5
GPIO_6
I2C_HOST_IRQ
I2C_HOST_SDA
I2C_HOST_SCL
GPIO_10
MOSI
SCK
MISO
GPIO_14
R8 10M
Y1
32.768kHz
C6 7pF
C7 7pF
GND
GND
3V3
GND
PSRAM
TXD0
RXD0
GPIO_17
GPIO_18
USB_D-
USB_D+
GPIO_21
GPIO_38
GPIO_39
GPIO_40
GPIO_41
GPIO_42
GPIO_46
GPIO_47
GPIO_48
UART0_TX
UART0_RX
USB D-
USB D+
GNSS
GPIO_42
GPIO_3
GPIO_48
R53 1k
PG_GNSS
PPS
EXTINT
P07
ANT_FAULT
I2C_HOST_SDA
I2C_HOST_SCL
JP1
JP2
SDA/SPI_CS
SCL/SPI_CLK
UART1_TX
R50 1k
UART1_RX
RXD/SPI_MOSI
TXD/SPI_MISO
gnss.kicad_sch
LoRa
GPIO_14
GPIO_4
GPIO_41
GPIO_40
R18 0
R19 0
BUSY
DIO1
EN
RESET
GPIO_10
SCK
MOSI
MISO
CS
SCK
MOSI
MISO
lora.kicad_sch
TP15
GPIO_EXP_RESET
Peripherals
GPIO_46
LED_DIN
RESET_IOEXP
P16
P17
P07
IO_P16
IO_P17
IO_P07
peripherals.kicad_sch
Connectors
GPIO_5
GPIO_38
GPIO_17
GPIO_18
GPIO_6
BUSY_DISPLAY
CS_BUS
CS_DISPLAY
D/C_DISPLAY
RESET_DISPLAY
connectors.kicad_sch
Rails
P16
P17
GPIO_39
GPIO_21
GPIO_47
PGOOD_3V3
PGOOD_5V
PGOOD_EFUSE
EN_BUCK_5V
EN_CLOCK
rails.kicad_sch
Pull up/down
+3V3
I2C_HOST_IRQ
I2C_HOST_SDA
I2C_HOST_SCL
R20 10k
R21 2.2k
R22 2.2k
JP3
JP4
JP5
C17 1uF
GND
FID4 Fiducial
FID5 Fiducial
FID6 Fiducial
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
H1
H2
H3
H4
GND
LimesKey
Sheet: /Root/
File: parsnip.kicad_sch
Title: ESP32-S3 MCU
Size: A4
Date: 2026-10-05
Rev:
KiCad E.D.A. 10.99.0-5499-g0f5acecd00
Id: 7/12


5V Buck
PGOOD_5V
TP11
PGOOD
VSYS
+3V3
U14
LM61460AASQRJRRQ1
VIN1
VIN2
PGND1
PGND2
PGOOD
BIAS
SW
CBOOT
RBOOT
FB
EN/SYNC
RT
VCC
AGND
SYNC_1
3 x 10uF 0603
C177
C189
C188
C145
2.2uF
10uF
22uF
5V_RAW
R119
10k
R120
10
3 x 22uF
C136
100nF
L5
3.3uH
R118
2.2
C139
220pF
R117
100
R121
402k
0.1%
R122
101k
0.1%
R124
1k
C158
5.6pF
C129
2.2uF
R107
402k
R108
100k
R110
6.04k
R103
100
R106
100k
Q15
2N7002
TP9
TP13
EN_BUCK_5V
GND
402k/101k at 0.1%, 4.980V nominal
Optional: eFuse for overvoltage protection,
reverse current blocking, and output slew rate control
U15
TPS259474ARPWR
IN
EN/UVLO
OVLO/OVCSEL
OUT
PGTH/FLT
PG/AUXOFF
ITIMER
DVDT
ILM
R125
86.6k
R126
3.92k
R127
27.4k
0.1%
C161
2.2nF
C162
2.2nF
R128
549
549 ohms is below
what's recommended but Idc
R129
280k
R131
100k
R130
100k
PGOOD_EFUSE
PWR_FLAG
TP14
D20
SMF6.0A
C168
10uF
C169
10uF
C166
22uF
C165
22uF
+5V
+3V3
5V_RAW
JP17
DISABLE_5V_PROT
3V3 Buck
PGOOD_3V3
TP10
PGOOD
U13
LM61460AASQRJRRQ1
C122
C123
C124
10uF
C125
2.2uF
R104
402k
R105
100k
R109
6.04k
TP8
TP7
EN_3V3
JP16
SYNC_2
2 x 10uF 0603
C141
C143
C192
C191
C190
R113
10k
C135
2.2uF
C131
100nF
L4
2.2uH
R112
2.2
C133
220pF
R111
100
R114
10
22uF 22uF
R115
402k
0.1%
R116
174k
0.1%
TP12
R123
1k
C153
5.6pF
PWR_FLAG
Optional: Buck's VIN 180 degrees out of
phase with eachother for less EMI
U16
SN74LVC1G74
SYNC_1
SYNC_2
TP6
TP16
C159
1nF
C160
1nF
C163
1uF
C164
1uF
R132
10k
D21
BAT54A
D22
BAT54W
EN_CLOCK
PGOOD_5V
PGOOD_3V3
Y2
SX1M4.000M10F30TNN
4MHz / 2 = 2MHz each buck
LimesKey
Sheet: /Root/Rails/
File: rails.kicad_sch
Title: 5 V & 3.3 V Buck Regulators
Size: A4
Date: 2026-10-05
Rev:
KiCad E.D.A. 10.99.0-5499-g0f5acecd00
Id: 8/12

LEDs and Jumpers

High to enable I2C/UART, low to enable SPI
Only for NEO-M[#]N

3V3_GNSS
D_SEL
JP11
R67 10k
R70 2k
D11 POWER
R136 100k
ANT_FAULT
D12 TIMEPULSE
PPS
R71 1k
GND
3V3_GNSS
JP10
TP5 CS_GPS
R68 2.2k
R72 2.2k
SDA/SPI_CS
SCL/SPI_CLK
GND

3V3_GNSS +3V3
V_BCKP
D14 ESD351DPYR
D15 RB531SM-30
R82 4.7k
PWR_FLAG
MS621FE 5.5mAh
BT3
C102 10uF
C103 10uF
C107 10uF
C99 10uF
C100 10uF
VCC_RF
U9 NEO-F10N
VCC
V_BCKP
VCC_RF
USB_DM/(F10:NC)
USB_DP/(F10:NC)
LNA_EN
RF_IN
GPS_ANT
TXD/SPI_MISO
RXD/SPI_MOSI
D_SEL/(F10:NC)
D_SEL
SDA/SPI_CS/(F10:NC)
SCL/SPI_CLK/(F10:NC)
TIMEPULSE
EXTINT
PPS
SW5 SAFEBOOT_GNSS
R76 10k
SAFEBOOT
RESET
RESET_GNSS SW4
VDD_USB
GND

Pin-Pin compatible with NEO-M9N/M8N/M7N/M6N & NEO-F10N
The NEO-F10N only supports UART

Sparkfun Schematic: https://cdn.sparkfun.com/assets/1/0/a/8/a/SparkFun_GPS_NEO_M9N_SMA_UPDATE.pdf
U-Blox Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_Integrationmanual_UBX-19014286.pdf

Optional LDO & eFuse

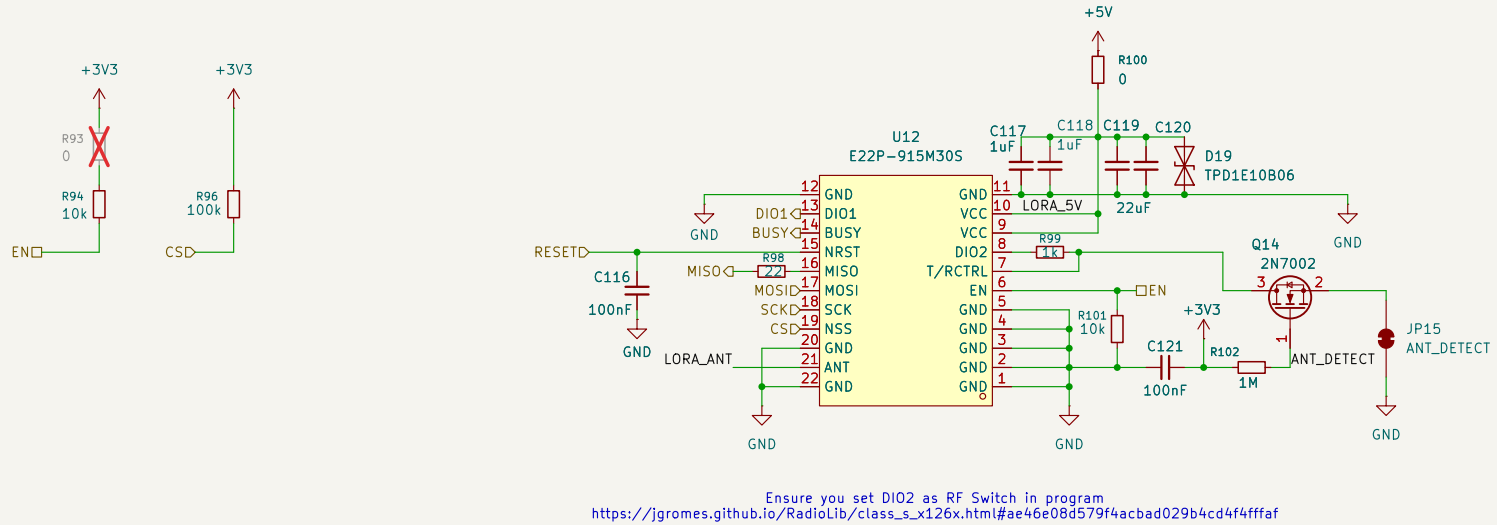
+5V
U10 TPS7A2033PDBVR
R134 0
IN
EN
GND
OUT
NC
C104 2.2uF
C130 10uF
High PSRR & Low Noise LDO

Optional module power input protection
U11 TPS259474LRPWR
R84 150k 0.1%
R85 215k 0.1%
R86 100k 0.1%
R87 113k 0.1%
EN/UVLO
OVLO/OVCSEL
ITIMER
DVDT
ILM
PGTH/FLT
PG/AUXOFF
C108 2.2nF
C109 2.2nF
R88 6.65k 1%
C106 10uF
D17 TPD1E10B06
3V3_GNSS
PWR_FLAG
R90 301k
R91 100k
R89 200k 0.1%
C110 10uF
C111 10uF
PG_GNSS
+3V3
JP13
DISABLE_GNSS_PROT
Overcurrent Threshold: 575mA

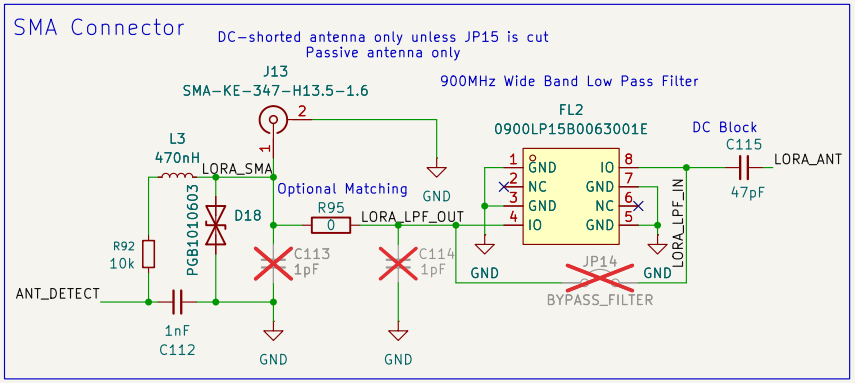
SMA + IPEX Connector

J11
Active Antenna only
J12
GNSS_SMA
GNSS_UFL
D13 ESD501
D10 ESD501
R69 0
R73 0
GNSS_BIAS
C95 47pF COG
Optional Matching
R75 0
C96 10nF
C97 10nF
Bandpass GNSS SAW filter
Compatible with L1/L5 GNSS bands
FL1 B39162B8389P810
Q=42 @ 1.8GHz
L6 5.1nH
L7 5.1nH
Attenuate 6dB
R81 37.4 1%
R80 150 0.5%
R83 150 0.5%
GPS_ANT
C93 22pF COG
L2 47nH
C94 1uF
R74 10
U6 TPS22945DCKR
Current limiting
~150mA trip
VOUT
VIN
ANT_FAULT
OC
GND
ON
R78 1k
LNA_EN
R79 100k
C138 1uF
C101 4.7uF
+5V
VCC_RF
D16 RB531SM-30
JP12
FB1 BLM18HG601SN1D

LimesKey	
Sheet: /Root/GNSS/	
File: gnss.kicad_sch	
Title: GNSS (NEO-F10N) & Active Antenna	
Size: A4	Date: 2026-09-14
KiCad E.D.A. 10.99.0-5499-g0f5acecd00	Rev: 1.1
	Id: 9/12



Ensure you set DIO2 as RF Switch in program
https://jgromes.github.io/RadioLib/class_s_x126x.html#ae46e08d579f4acbad029b4cd4f4fffaf



LimesKey

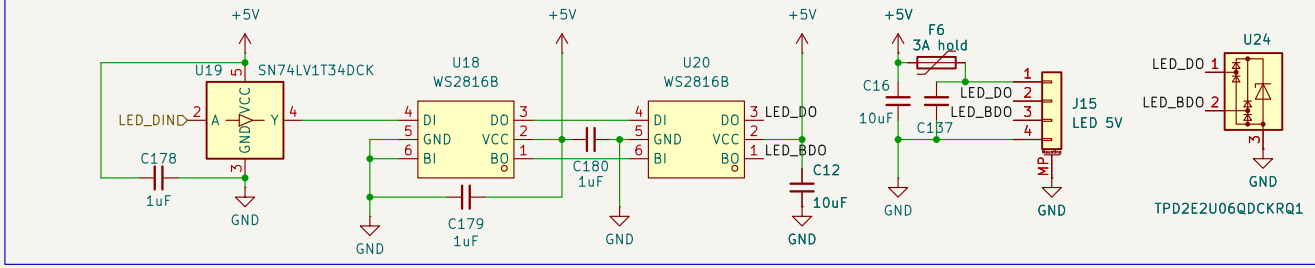
Sheet: /Root/LoRa/
File: lora.kicad_sch

Title: LoRa (E22P-915M30S) & Antenna

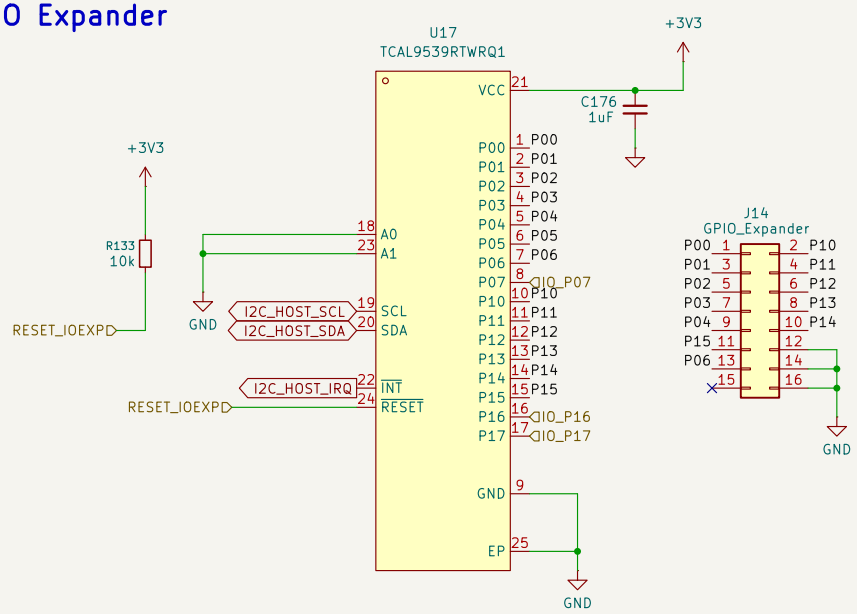
Size: A4	Date: 2026-08-24	**Rev:**
KiCad E.D.A. 10.99.0-5499-g0f5acecd00		Id: 10/12

Individually Addressible RGB LEDs

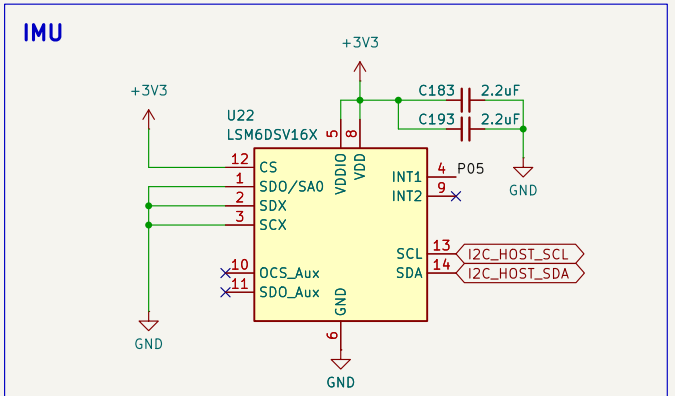
For use with WS2813/WS2816



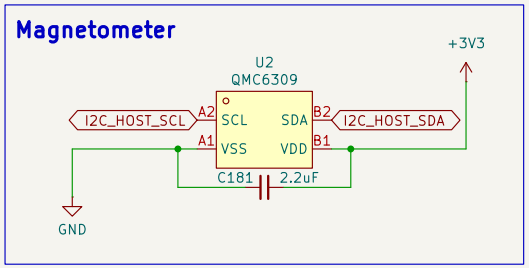
GPIO Expander



IMU

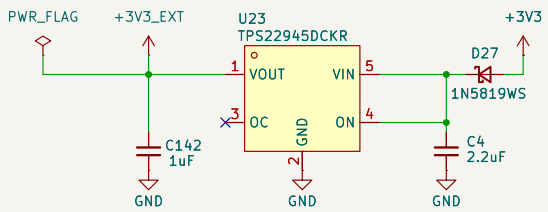


Magnetometer



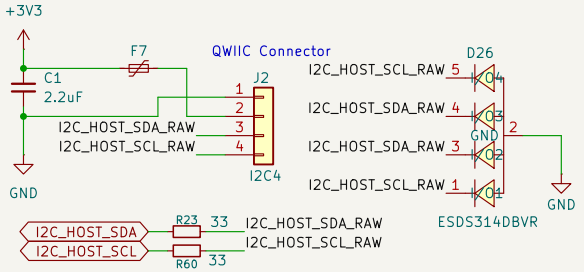
LimesKey		
Sheet: /Root/Peripherals/		
File: peripherals.kicad_sch		
Title: GPIO Expander, RGB LEDs, Magnetometer		
Size: A4	Date: 2026-10-05	**Rev:**
KiCad E.D.A. 10.99.0-5499-g0f5acecd00		Id: 11/12

Low-input-voltage Current-limited Load Switches With Shut Off And Auto-Restart Feature

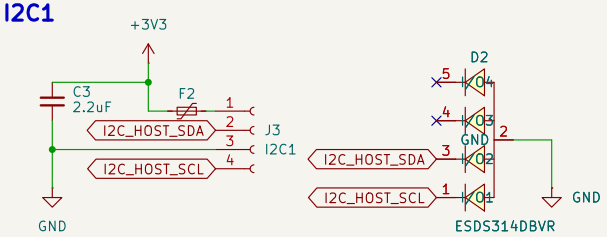


U23: Cin capacitance must be greater than Cout capacitance (locally)

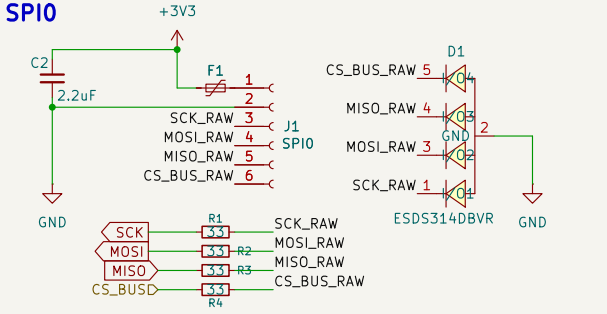
QWIIC I2C4



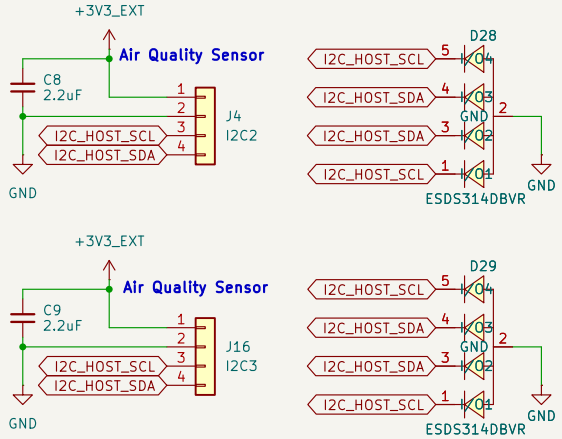
I2C1



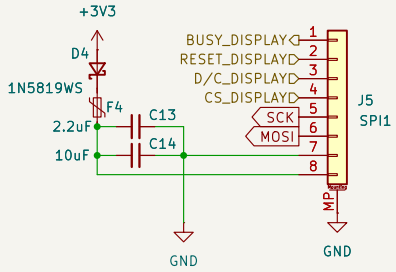
SPI0



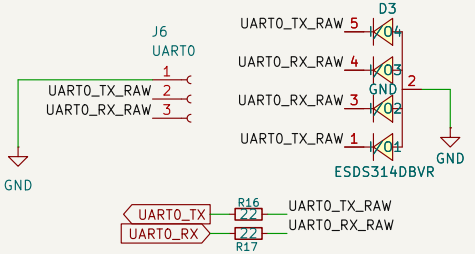
I2C2 & I2C3



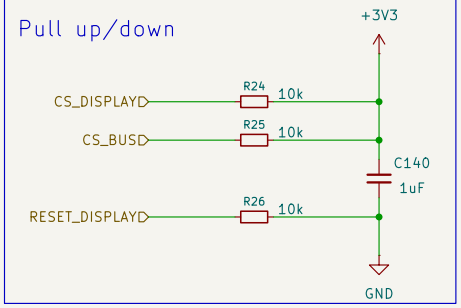
ePaper Display SPI1



UART0



Pull up/down



LimesKey	
Sheet: /Root/Connectors/	
File: connectors.kicad_sch	
Title: Expansion & Display Connectors	
Size: A4	Date: 2026-10-05
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